

Week 1 facilitator guide: One from every pile

Bulgarian solitaire for grades K–5, three tables, one hour

The session at a glance

The game. Counters sit in piles. The only move: take one counter from every pile and put all the counters you took together as one new pile; a pile that reaches zero is gone. The arrangement is the list of pile sizes, written biggest first (3 2 1). The children run this deterministic process by hand and record it.

The mathematics (what the tables are really finding out). With a triangular number of counters (1, 3, 6, 10, 15, ...) every start ends at the staircase $(k, k - 1, \dots, 1)$, which the move does not change (Brandt, 1982). With any other number of counters there is no stop at all and every start falls into a loop. On the way the children meet: which arrangements are fixed (only staircases), completeness (“all eleven ways” and “all seven ways”), the longest run to the stop (2, 6, 12, 20 moves for 3, 6, 10, 15 counters), arrangements that nothing leads to (Garden of Eden states, with a clean test), and loops that are “stairs plus an extra counter walking down the steps.”

Timeline (60 minutes).

- 0–4: children arrive, sit at their tables and handle the counters freely (stacking, sorting, making piles). Packets stay face down.
- 4–7: whole-group launch at one table or on the floor (script below).
- 7–50: table work in the order given for each table, about 40 minutes of real working time.
- 50–58: closing share (script below).
- 58–60: counters back in cups, packets go home with the children.

Who sits where. Recommended: the organizer (math PhD) with grades 4–5, where the arguments are longest; the second mathematician with K–1, where the adult reads everything aloud, scribes, and has to keep the real questions alive (“which starts stop? all of them?”) for three young children; the parent with the four third graders, who read their own pages and can work in pairs, with every answer and hint printed in this guide. If the parent is the one most at ease with five-year-olds, swap the K–1 and 2–3 adults; the K–1 section below is fully scripted either way.

What each table has. K–1: 8 pages, Problems 1–15, boards of circles to color; the adult reads. Grades 2–3: 6 pages, Problems 1–15, strips of boxes to write pile sizes in. Grades 4–5: 8 pages, Problems 1–14. Nobody is expected to finish every problem; each section says which problems are the core and which are for a table that is ahead.

Materials and preparation

- Counters: 30 per child in a cup (300 in all), plus 10 for the launch. Flat counters that stack are best, since a pile can be built as a stack and “take one from every pile” is then “take the top of every stack.” Any one kind of object will do; if counters run short, face-down playing cards are counters too.
- A tray or dark cloth for the launch so that the piles are visible from all three tables.

- Pencils for everyone; for the K–1 table also colored pencils or thin markers in two colors (Problem 14 uses two colors). Crayons are too fat for the circles.
- Packets, stapled: K–1 pages, one per child plus one spare (4 copies); grades 2–3 pages (5 copies); grades 4–5 pages (4 copies); this guide (3 copies, one per adult).
- Before the session, each adult plays the game once with counters from 6 (it stops at 3 2 1 after 3 moves) and once from 5 (it loops: 3 2, 2 2 1, 3 1 1). Two minutes, and the move is then in your hands.
- Each adult reads this guide’s section for their own table and the reference tables at the end, which list every arrangement of 4, 5, 6, 7, 8 and 10 counters with the arrangement it becomes, the number of moves to the stop or loop, and the arrangements that lead to it. Nothing needs to be worked out during the hour.

Words used on the pages

Use the same words at every table so that the closing share makes sense to everyone.

- **the move**: take one counter from every pile; the counters you took become one new pile.
- **arrangement**: the pile sizes in order, biggest first, written 3 2 1. Two arrangements with the same sizes in a different order are the same arrangement. (K–1 says “picture” instead: each pile is colored as a tower of circles.)
- **stop**: an arrangement that the move does not change (2 1, 3 2 1, 4 3 2 1, ...). K–1 says “the picture stays the same.”
- **loop**: a set of arrangements that the move cycles through again and again.
- **stairs**: piles 1, 2, 3, ..., one of each size. “Perfect stairs” on the K–1 pages means no counter left over.
- **towers of dots**: an arrangement drawn with one column of dots per pile. This is the Young diagram of the partition, and it is the picture in which the arguments are easiest.

Whole-group launch (three minutes)

Everyone gathers around the tray. The organizer has 7 counters in one pile.

“This is the only move in today’s game. I take one counter from every pile, and the counters I took make a new pile.” Do it: the single pile of 7 loses one, and the one counter taken sits alone as a new pile. “Now the piles are 6 and 1. Say the move with me: one from every pile, make a new pile.” Do it with the group chanting: take one from the 6 and one from the 1, which is now gone, and put the two together: 5 and 2. Again: one from the 5, one from the 2, the two taken make a new pile: 4 2 1. Stop there.

“Three questions for today. Does the game ever stop changing? Does it go around in a circle? Does it matter how you start? Each table has its own numbers of counters to find out with. Your first problem tells you where to begin.”

Do not continue the demonstration past 4 2 1: with 7 counters the game never stops, and the tables should discover stops and loops themselves. (If a child asks what happens next with 7, say “that is on one of your pages.” It is on every table’s pages.)

Table K–1 (two kindergartners, one first grader)

How the table runs. Read each problem aloud as written, one sentence at a time, and show the first move with your own counters. Each child has a packet and a cup of counters. Children build piles on the table, in a row from biggest to smallest, and record by coloring: one tower of circles per pile, as tall as the pile, from the bottom line up. Say the sizes aloud as they color (“a tower of 3, a tower of 2, a tower of 1”), and write the numbers under the towers yourself when that helps. A child who colors the wrong number of circles recounts the real pile; the counters are the truth, the picture is the record.

Order and timing (40 minutes). Core: Problems 1–2 (8 min), 3 (4 min), 5–6 (8 min), 7–8 (6 min), 10–11 (7 min), 12 (4 min). Then, in this order, as time allows: 4, 9, 13, 14, 15. A table that is slow can skip Problem 4 and give Problem 8 to one child only while the others do Problem 9.

Problem 1. The three ways to make piles with 3 counters are 3, 2 1 and 1 1 1. Have everyone do row 1 together starting from 1 1 1: $1\ 1\ 1 \rightarrow 3 \rightarrow 2\ 1 \rightarrow 2\ 1$ (pictures 1–4; the fourth picture repeats the third, which is the signal to stop). Row 2 from 3: $3 \rightarrow 2\ 1 \rightarrow 2\ 1$ (three pictures). Row 3 from 2 1: $2\ 1 \rightarrow 2\ 1$ (two pictures). Taking one from a pile of 1 makes that pile disappear; show it with the counters and say “that pile is gone.”

Problem 2. Every row ends with the stairs 2 1, and the move leaves 2 1 alone: the two counters taken make a new pile of 2, the old 2 becomes 1, the old 1 disappears. Three times in a row, nothing changes. Say: “With 3 counters, every start ends at the stairs, and the stairs stay.” That sentence is the whole theorem in small.

Problem 3. $4 \rightarrow 3\ 1 \rightarrow 2\ 2 \rightarrow 2\ 1\ 1 \rightarrow 3\ 1 \rightarrow 2\ 2$. Pictures 2 and 5 match (3 1), and so do 3 and 6 (2 2). Nothing ever stays the same: with 4 counters the game goes around a loop of three arrangements, 3 1, 2 2, 2 1 1. Ask: “Will it ever make stairs?” and let a child run it a few more times with counters: it never does.

Problem 4. $1\ 1\ 1\ 1 \rightarrow 4 \rightarrow 3\ 1 \rightarrow 2\ 2 \rightarrow 2\ 1\ 1 \rightarrow 3\ 1$. Pictures 3 and 6 match. Pictures 2 to 6 all appear in Problem 3 (4 is Problem 3’s picture 1), so a check mark goes under each of them. The point to say aloud: “a different start, and after one move it is the same game as before.”

Problem 5. $6 \rightarrow 5\ 1 \rightarrow 4\ 2 \rightarrow 3\ 2\ 1 \rightarrow 3\ 2\ 1$. Pictures 4 and 5 match; 3 arrows, so the number in the box is 3. The stairs with three steps stay. Ask: “Is that the only way to start with 6? What else could we try?” This leads into Problems 7–9.

Problem 6. Crossing out the top circle of each tower leaves towers of 2, 1 and 0; the three crossed-out circles make a new tower of 3 in the empty space. The towers are now 2, 1, 3: the same three sizes in a different order. With the real counters lined up biggest first, the piles read 3 2 1 again. The reason, in words a child can repeat: “There are three piles, so the new pile has 3. The old 3 becomes 2, the old 2 becomes 1, the old 1 is gone. Three, two, one, the same as before.” Ask “Would it work for stairs with four steps?” and keep that for Problem 11.

Problem 7. $1\ 1\ 1\ 1\ 1\ 1 \rightarrow 6 \rightarrow 5\ 1$. Picture 3 is 5 1, which is picture 2 of Problem 5. What happens next is what happened there: 4 2, then 3 2 1, then it stays; 4 moves in all from the six piles of 1. The idea to say aloud: once two starts reach the same picture, they go the same way from then on.

Problem 8. $3\ 3 \rightarrow 2\ 2\ 2 \rightarrow 3\ 1\ 1\ 1 \rightarrow 4\ 2 \rightarrow 3\ 2\ 1 \rightarrow 3\ 2\ 1$. Pictures 5 and 6 match; 4 arrows. Ask: “Have we seen 4 2 before?” (Problem 5, picture 3.)

Problem 9. Every one of the 11 ways to make piles with 6 counters ends at 3 2 1. The ways not yet used at the table, with the number of moves: 5 1 (2), 4 2 (1), 4 1 1 (5), 3 1 1 1 (2), 2 2 2 (3), 2 1 1 1 1 (3), 2 2 1 1 (6). The page has 9 pictures, enough for any of them. Offer 2 2 1 1 to a child who wants a long one: $2\ 2\ 1\ 1 \rightarrow 4\ 1\ 1 \rightarrow 3\ 3 \rightarrow 2\ 2\ 2 \rightarrow 3\ 1\ 1\ 1 \rightarrow 4\ 2 \rightarrow 3\ 2\ 1 \rightarrow 3\ 2\ 1$, 6 moves, the longest possible

with 6 counters. Full list in the reference table for 6 counters.

Problem 10. $5 \rightarrow 4 \ 1 \rightarrow 3 \ 2 \rightarrow 2 \ 2 \ 1 \rightarrow 3 \ 1 \ 1 \rightarrow 3 \ 2$. Pictures 3 and 6 match; the game loops through $3 \ 2, 2 \ 2 \ 1, 3 \ 1 \ 1$. No picture is a stairs shape, so no star gets drawn. Compare with Problem 5 aloud: “6 counters stop at the stairs; 5 counters never stop. Why could that be?” A good child answer: “5 counters cannot make stairs, there is one too few or two too many.” Problem 12 makes that precise.

Problem 11. The stairs $4 \ 3 \ 2 \ 1$ use 10 counters. The move takes one from each of four piles, so the new pile has 4; the old piles become 3, 2, 1 and nothing: $4 \ 3 \ 2 \ 1$ again. The picture after the move is the same stairs. Repeat the Problem 6 words with four piles.

Problem 12. Perfect stairs: 1 (a single pile of 1), 3 ($2 \ 1$), 6 ($3 \ 2 \ 1$), 10 ($4 \ 3 \ 2 \ 1$). Circle 1, 3, 6, 10; cross out 2, 4, 5, 7, 8, 9, 11, 12. With 7 counters: stairs $3 \ 2 \ 1$ and one counter left over, and the next step needs 4 counters, so 8 and 9 fail too; 10 works; 11 and 12 leave counters over. Say the sequence aloud: “1, 3, 6, 10. Each time we add the next step: plus 2, plus 3, plus 4.”

Problem 13. Five steps: 15 counters ($10 + 5$). Six steps: 21 ($15 + 6$).

Problem 14. $4 \ 2 \ 1 \rightarrow 3 \ 3 \ 1 \rightarrow 3 \ 2 \ 2 \rightarrow 3 \ 2 \ 1 \ 1 \rightarrow 4 \ 2 \ 1$. Pictures 1 and 5 match, a loop of four. The extra counter (the one beyond the stairs $3 \ 2 \ 1$) sits on top of pile 1, then on top of pile 2, then on top of pile 3, then it is a new fourth pile of 1, then it is back on pile 1: it walks down the stairs one step per move and jumps back to the top. In the pictures: the stairs in one color stay in place and the other color moves one tower to the right each time.

Problem 15. $5 \ 3 \ 2 \ 1 \rightarrow 4 \ 4 \ 2 \ 1 \rightarrow 4 \ 3 \ 3 \ 1 \rightarrow 4 \ 3 \ 2 \ 2 \rightarrow 4 \ 3 \ 2 \ 1 \ 1 \rightarrow 5 \ 3 \ 2 \ 1$. Pictures 1 and 6 match; 5 arrows between them, a loop of five. The same walk as in Problem 14, down four steps and back.

Hints, in the order to give them. (1) “Touch each pile and take one; hold the taken ones in your other hand; now put them down together, that is the new pile.” (2) “Line the piles up, biggest first, before you color.” (3) For spotting a repeat: build the two candidate pictures side by side with counters. (4) For “why does it stay” (Problems 6 and 11): “How many piles are there? So how many counters are in the new pile?” (5) For Problem 12: “Make the stairs as far as you can; what is left over? What would the next step need?”

If the table stalls. If the move itself is shaky, do Problem 1 row 1 again as a group with the chant. If coloring is slow, let the child color only the start and the final picture of a run and say the middle ones aloud while you write the numbers. If a child is upset that 4 or 5 counters “never finish,” say that the game has two kinds of endings, stopping and going around, and that finding which happens is the job.

If the table finishes early. Problems 4, 9, 13, 14, 15 in that order. After that: 8 counters from 8 loops through $4 \ 3 \ 1, 3 \ 3 \ 2, 3 \ 2 \ 2 \ 1, 4 \ 2 \ 1 \ 1$ (a loop of four, reached after 4 moves); and the challenge “find a start with 6 counters that takes more than 4 moves to reach the stairs” ($4 \ 1 \ 1$ takes 5, $2 \ 2 \ 1 \ 1$ takes 6).

Table grades 2–3 (four third graders)

How the table runs. The children read their own pages and work in two pairs, each pair with one set of counters and both packets being filled in. Pile sizes are written biggest first in the boxes; the first box of a strip is pre-filled when the start is given. When a strip is done, ask the pair to read it aloud to the other pair.

Order and timing (40 minutes). Core: Problems 1–4 (8 min), 5 (2 min), 6–7 (7 min), 8–9 (10 min), 10 (5 min), 11 (4 min). Then 12–13 (5 min), 14 (4 min), 15 (extra). If the table is slow, skip Problems 12–13 and go to 14, which uses the strips of Problem 8.

Problem 1. $4 \rightarrow 3\ 1 \rightarrow 2\ 2 \rightarrow 2\ 1\ 1 \rightarrow 3\ 1$. Boxes 2 and 5 match.

Problem 2. $1\ 1\ 1\ 1 \rightarrow 4 \rightarrow 3\ 1 \rightarrow 2\ 2 \rightarrow 2\ 1\ 1 \rightarrow 3\ 1$ (boxes 3 and 6). $2\ 2 \rightarrow 2\ 1\ 1 \rightarrow 3\ 1 \rightarrow 2\ 2$ (boxes 1 and 4).

Problem 3. The two remaining ways are $3\ 1$ and $2\ 1\ 1$: $3\ 1 \rightarrow 2\ 2 \rightarrow 2\ 1\ 1 \rightarrow 3\ 1$, and $2\ 1\ 1 \rightarrow 3\ 1 \rightarrow 2\ 2 \rightarrow 2\ 1\ 1$. If a pair cannot find them, ask “what is the biggest pile you have not tried?” (3, then 2 with the rest in ones).

Problem 4. The loop is $3\ 1 \rightarrow 2\ 2 \rightarrow 2\ 1\ 1 \rightarrow$ back to $3\ 1$; three arrangements. All five starts fall into it, two of them (4 and $1\ 1\ 1\ 1$) after a lead-in. With 4 counters there is no stop at all.

Problem 5. $6 \rightarrow 5\ 1 \rightarrow 4\ 2 \rightarrow 3\ 2\ 1 \rightarrow 3\ 2\ 1$. The stop is $3\ 2\ 1$.

Problem 6. There are 11 ways: 6, 5 1, 4 2, 4 1 1, 3 3, 3 2 1, 3 1 1 1, 2 2 2, 2 2 1 1, 2 1 1 1 1, 1 1 1 1 1 1. Most pairs find 8 to 10 at first; do not correct them, Problem 7 does.

Problem 7. Rows of the table: biggest pile 6: 6. Biggest 5: 5 1. Biggest 4: 4 2, 4 1 1. Biggest 3: 3 3, 3 2 1, 3 1 1 1. Biggest 2: 2 2 2, 2 2 1 1, 2 1 1 1 1. Biggest 1: 1 1 1 1 1 1. Total 11. The completeness argument in a child’s words: “if the biggest pile is 4, the other 2 counters can only be 2 or 1 1; if the biggest pile is 3, the other 3 counters can be 3, 2 1 or 1 1 1, and nothing bigger than 3 is allowed; if the biggest is 2, the other 4 are 2 2, 2 1 1 or 1 1 1 1.” Row by row there is nowhere for a missing way to hide.

Problem 8. Every run ends at $3\ 2\ 1$. The number of moves for each start: 6: 3; 5 1: 2; 4 2: 1; 4 1 1: 5; 3 3: 4; 3 2 1: 0; 3 1 1 1: 2; 2 2 2: 3; 2 2 1 1: 6; 2 1 1 1 1: 3; 1 1 1 1 1 1: 4. The full runs are in the reference table for 6 counters; the longest is $2\ 2\ 1\ 1 \rightarrow 4\ 1\ 1 \rightarrow 3\ 3 \rightarrow 2\ 2\ 2 \rightarrow 3\ 1\ 1\ 1 \rightarrow 4\ 2 \rightarrow 3\ 2\ 1$. Split the eleven starts between the pairs (six and five).

Problem 9. Yes, every start reaches the same stop $3\ 2\ 1$. Most moves: $2\ 2\ 1\ 1$, 6 moves. Fewest apart from the stop itself: 4 2, 1 move. Arrangements that appear in more than one strip: 4 2 is in every strip except the one that starts at the stop; 5 1 is in the runs from 6, $1\ 1\ 1\ 1\ 1\ 1$ and $2\ 1\ 1\ 1\ 1$; 6 is in the run from $1\ 1\ 1\ 1\ 1\ 1$; the run from $2\ 2\ 1\ 1$ passes through 4 1 1, 3 3, 2 2 2 and 3 1 1 1, and the runs from those four starts are its tail ends. The question behind this: once two runs meet, they are the same run from then on, so all eleven runs are really one tree of paths flowing into $3\ 2\ 1$.

Problem 10. Crossing out the top dot of each tower leaves 2, 1, 0 and the three crossed-out dots make a new tower of 3; sorted, the towers are 3, 2, 1 again. Why the move can never change $3\ 2\ 1$: there are three piles, so the new pile has exactly 3; the old piles 3, 2, 1 become 2, 1 and nothing. The sizes after the move are 3, 2, 1. For $4\ 3\ 2\ 1$: four piles, so the new pile has 4; the old piles become 3, 2, 1, nothing; the sizes are 4, 3, 2, 1 again. With 10 counters the move indeed changes nothing. Ask “would the same reason work for five steps?” and accept “five piles make a new pile of 5, and the others go down to 4, 3, 2, 1.”

Problem 11. 1 2: 3 counters, a stop. 1 2 3: 6, a stop. Up to 4: 10, a stop. Up to 5: 15, a stop. Up to 6: 21, a stop. The numbers 3, 6, 10, 15, 21 go up by 3, 4, 5, 6: each new step adds its own size.

Problem 12. The 7 ways and what one move does to each: $5 \rightarrow 4\ 1$; $4\ 1 \rightarrow 3\ 2$; $3\ 2 \rightarrow 2\ 2\ 1$; $3\ 1\ 1 \rightarrow 3\ 2$; $2\ 2\ 1 \rightarrow 3\ 1\ 1$; $2\ 1\ 1\ 1 \rightarrow 4\ 1$; $1\ 1\ 1\ 1\ 1 \rightarrow 5$. No right box equals its left box, so nothing gets circled: with 5 counters there is no stop. This is a complete-case proof, and worth saying so: “we did not find a stop, and we also know there is none, because we checked all seven.”

Problem 13. $5 \rightarrow 4\ 1 \rightarrow 3\ 2 \rightarrow 2\ 2\ 1 \rightarrow 3\ 1\ 1 \rightarrow 3\ 2$ (boxes 3 and 6). $2\ 1\ 1\ 1 \rightarrow 4\ 1 \rightarrow 3\ 2 \rightarrow 2\ 2\ 1 \rightarrow 3\ 1\ 1 \rightarrow 3\ 2$. The loop: $3\ 2 \rightarrow 2\ 2\ 1 \rightarrow 3\ 1\ 1 \rightarrow$ back to $3\ 2$; three arrangements. All seven starts end in this loop.

Problem 14. Gives 3 3 after one move: 4 1 1, and that is the only one. Gives 4 2: 5 1 and 3 1 1 1. Gives $2\ 2\ 1\ 1$: nothing. The children can see this from Problem 8, since $2\ 2\ 1\ 1$ never appears after an arrow in

any of the eleven strips, and every arrangement of 6 counters has a strip. The three arrangements that never appear after an arrow: 2 2 1 1, 2 1 1 1 1, 1 1 1 1 1 1. The reason, if a pair wants one: after a move, the new pile has as many counters as there were piles before the move, and the number of piles can grow by at most one (the new pile is added, some old piles may vanish). So after a move the biggest pile is at least the number of piles minus one. In 2 2 1 1 the biggest pile is 2 but there are 4 piles.

Problem 15. $7 \rightarrow 6\ 1 \rightarrow 5\ 2 \rightarrow 4\ 2\ 1 \rightarrow 3\ 3\ 1 \rightarrow 3\ 2\ 2 \rightarrow 3\ 2\ 1\ 1 \rightarrow 4\ 2\ 1$: a loop of four. $8 \rightarrow 7\ 1 \rightarrow 6\ 2 \rightarrow 5\ 2\ 1 \rightarrow 4\ 3\ 1 \rightarrow 3\ 3\ 2 \rightarrow 3\ 2\ 2\ 1 \rightarrow 4\ 2\ 1\ 1 \rightarrow 4\ 3\ 1$: a loop of four. $5\ 3 \rightarrow 4\ 2\ 2 \rightarrow 3\ 3\ 1\ 1 \rightarrow 4\ 2\ 2$: a loop of two. So 8 counters have two different loops, and where you end depends on where you start; 4, 5, 6 and 7 counters each have only one ending.

Hints, in the order to give them. (1) “Say the move as you do it: one from every pile, make a new pile.” (2) “Sort biggest first before you write.” (3) For Problem 6: “What is the biggest pile you have not used yet?” (4) For Problem 10: “How many piles are there? Then how many counters go into the new pile? What happens to each old pile?” (5) For Problem 14: “After a move, how big is the new pile compared with the number of piles?”

If the table stalls. On Problem 6, let one pair do the table of Problem 7 straight away and report. On Problem 8, reduce to six starts (6, 3 3, 2 2 2, 4 1 1, 2 2 1 1, 1 1 1 1 1 1), which still show the longest run and the merging paths. On Problem 10, do it with counters first, pictures second.

If the table finishes early. Problem 15, then: “Draw the map of 6 counters: write all eleven arrangements on a blank sheet and draw an arrow from each to the one it becomes” (the map is drawn in the grades 4–5 section); then “find the loop for 9 counters” (4 3 2, 3 3 2 1, 4 2 2 1, 4 3 1 1, a loop of four, from 9 after 5 moves); then “find a start with 10 counters that takes more than 6 moves to reach 4 3 2 1” (reference table for 10 counters; 3 3 2 1 1 takes 12).

Table grades 4–5 (two fourth graders, one fifth grader)

How the table runs. The three work as a group on one set of counters, with each child recording in their own packet, and they share out the eleven runs of Problem 4. Push for the arguments in Problems 6, 7 and 8: the children should be able to say each one in their own words at the closing share.

Order and timing (40 minutes). Core: Problems 1 (5 min), 2 (2), 3 (5), 4 (7), 5 (6), 6 (4), 7 (6), 8 (5). Then 9 (5), 10 (6), 11 (6), 12 (4), 13 (8), 14 (10) as time allows. A table that is slow can give the map (Problem 5) a shorter form: only the chain from 2 2 1 1 and the arrows into 4 2.

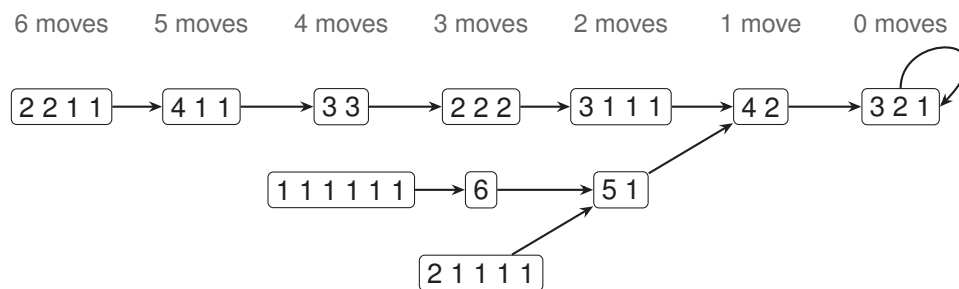
Problem 1. $4 \rightarrow 3\ 1 \rightarrow 2\ 2 \rightarrow 2\ 1\ 1 \rightarrow 3\ 1$ (boxes 2 and 5). $2\ 2 \rightarrow 2\ 1\ 1 \rightarrow 3\ 1 \rightarrow 2\ 2$ (boxes 1 and 4). $1\ 1\ 1\ 1 \rightarrow 4 \rightarrow 3\ 1 \rightarrow 2\ 2 \rightarrow 2\ 1\ 1 \rightarrow 3\ 1$ (boxes 3 and 6). The loop: $3\ 1 \rightarrow 2\ 2 \rightarrow 2\ 1\ 1 \rightarrow$ back; 3 arrangements. The two remaining starts, 3 1 and 2 1 1, are in the loop already, so all five starts end in it and there is no stop with 4 counters.

Problem 2. $6 \rightarrow 5\ 1 \rightarrow 4\ 2 \rightarrow 3\ 2\ 1 \rightarrow 3\ 2\ 1$; the stop is 3 2 1.

Problem 3. Eleven ways: 6; 5 1; 4 2, 4 1 1; 3 3, 3 2 1, 3 1 1 1; 2 2 2, 2 2 1 1, 2 1 1 1 1; 1 1 1 1 1 1. The organization to aim for: by the size of the biggest pile, and within that, by the size of the second pile (“then the rest can only be split in these ways”). Any systematic order that the child can explain is a complete answer; what matters is that they can say why nothing is missing.

Problem 4. Moves to the stop: 6: 3; 5 1: 2; 4 2: 1; 4 1 1: 5; 3 3: 4; 3 2 1: 0; 3 1 1 1: 2; 2 2 2: 3; 2 2 1 1: 6; 2 1 1 1 1: 3; 1 1 1 1 1 1: 4. Full runs in the reference table.

Problem 5. The map of 6 counters, with the number of moves to the stop as the column:



(a) No arrow points to 2 2 1 1, 2 1 1 1 1, 1 1 1 1 1 1. (b) Two arrows point to 4 2 (from 5 1 and 3 1 1 1); two to 5 1 (from 6 and 2 1 1 1 1); two to 3 2 1 (from 4 2 and from itself). (c) The longest chain: 2 2 1 1 → 4 1 1 → 3 3 → 2 2 2 → 3 1 1 1 → 4 2 → 3 2 1, six moves. Things to notice with the children: the map has no loop except the stop's own arrow, every path leads to 3 2 1, and every path passes through 4 2.

Problem 6. Crossing out the top dots of 3 2 1 leaves 2, 1, 0; the three dots make a new tower of 3; sorted, 3 2 1 again. The argument: three piles, so the new pile is 3; the old piles 3, 2, 1 become 2, 1, gone. For 4 3 2 1: four piles make a new pile of 4; the old ones become 3, 2, 1, gone. For 5 4 3 2 1 with 15 counters: five piles make a new pile of 5, the old ones become 4, 3, 2, 1, gone. The same reason works for stairs of any height: the number of piles equals the biggest pile, and each pile stepping down by one produces exactly the next pile's size.

Problem 7. (a) 3, one from each of the three piles. (b) The new pile. Every old pile lost a counter, so no old pile is as big as the old biggest pile was; but the arrangement after the move is the same as before, so some pile has the old biggest size, and it can only be the new pile. (c) The biggest pile of the stop is 3. (d) After the move, the second biggest pile is the old biggest pile minus one (the other old piles minus one are no bigger than it, and the new pile is the biggest): $3 - 1 = 2$. So the second biggest pile of the stop is 2. Continuing, the third biggest after the move is the old second biggest minus one, $2 - 1 = 1$, so the third pile is 1. (e) 3 2 1. (f) With 4 piles the new pile is 4 and the matching gives 4 3 2 1; with 5 piles, 5 4 3 2 1. A stop with p piles can only be the stairs with p steps. A detail to check with the children: the smallest pile of a stop must be 1, because if no pile were 1, no pile would vanish and the move would make $p + 1$ piles out of p . (g) 1, 3, 6, 10, 15, 21, 28. So the only numbers of counters that have a stop are the stairs numbers; with any other number the game never stops, and since there are finitely many arrangements it must repeat, that is, loop.

Problem 8. (a) 4 1 1. (b) 5 1 and 3 1 1 1, no others. (c) None. Reason: after a move, the new pile has as many counters as there were piles before the move, and the number of piles after the move is at most one more than before (one pile is added; piles of 1 vanish). So after any move, the biggest pile is at least the number of piles minus one. 2 2 1 1 has 4 piles and biggest pile 2, and $2 < 3$. (d) 6: 1 (from 1 1 1 1 1 1); 5 1: 2; 2 2 2: 1 (from 3 3); 3 2 1: 2 (from 4 2 and from itself). (e) The three with no incoming arrow: 2 2 1 1 (biggest 2, 4 piles), 2 1 1 1 1 (2, 5 piles), 1 1 1 1 1 1 (1, 6 piles): in each, the biggest pile is at least 2 less than the number of piles. Any three with an incoming arrow, for instance 4 2 (4, 2 piles), 2 2 2 (2, 3 piles), 3 1 1 1 (3, 4 piles): the biggest pile is at least the number of piles minus one. The test: an arrangement can be reached by a move exactly when its biggest pile is at least its number of piles minus one. The "can be reached" half comes with a recipe: take away a pile whose size m is at least the number of piles minus one, add one counter to every other pile, and add piles of 1 until there are m piles; one move from that arrangement gives the original. For 3 3: take away a 3, the other pile becomes 4, add two piles of 1 to make three piles: 4 1 1.

Problem 9. Prediction by the test: nothing becomes 3 1 1 1 1 (biggest 3, 5 piles, and $3 < 4$), 2 2 2 1 (2, 4 piles), 2 2 1 1 1 (2, 5 piles), 2 1 1 1 1 1 (2, 6 piles), 1 1 1 1 1 1 1 (1, 7 piles): five of the fifteen. Each of the other ten is reached: 7 from 1 1 1 1 1 1 1; 6 1 from 7 or 2 1 1 1 1 1; 5 2 from 6 1 or 3 1 1 1 1; 5 1 1 from 2 2 1 1 1; 4 3 from 5 1 1 or 4 1 1 1; 4 2 1 from 5 2 or 3 2 1 1; 4 1 1 1 from 2 2 2 1; 3 3 1 from 4 2 1;

3 2 2 from 4 3 or 3 3 1; 3 2 1 1 from 3 2 2.

Problem 10. 5 counters: $5 \rightarrow 4\ 1 \rightarrow 3\ 2 \rightarrow 2\ 2\ 1 \rightarrow 3\ 1\ 1 \rightarrow 3\ 2$; $2\ 2\ 1 \rightarrow 3\ 1\ 1 \rightarrow 3\ 2 \rightarrow 2\ 2\ 1$; the loop 3 2, 2 2 1, 3 1 1 (3). 7 counters: $7 \rightarrow 6\ 1 \rightarrow 5\ 2 \rightarrow 4\ 2\ 1 \rightarrow 3\ 3\ 1 \rightarrow 3\ 2\ 2 \rightarrow 3\ 2\ 1\ 1 \rightarrow 4\ 2\ 1$; the loop 4 2 1, 3 3 1, 3 2 2, 3 2 1 1 (4). 8 counters: $8 \rightarrow 7\ 1 \rightarrow 6\ 2 \rightarrow 5\ 2\ 1 \rightarrow 4\ 3\ 1 \rightarrow 3\ 3\ 2 \rightarrow 3\ 2\ 2\ 1 \rightarrow 4\ 2\ 1\ 1 \rightarrow 4\ 3\ 1$; the loop 4 3 1, 3 3 2, 3 2 2 1, 4 2 1 1 (4). $5\ 3 \rightarrow 4\ 2\ 2 \rightarrow 3\ 3\ 1\ 1 \rightarrow 4\ 2\ 2$; the loop 4 2 2, 3 3 1 1 (2). Eight counters have two loops; 7 of the 22 arrangements of 8 counters end in the short one (6 1 1, 5 3, 4 2 2, 3 3 1 1, 4 1 1 1 1, 2 2 2 2, 2 2 1 1 1 1).

Problem 11. The loop for 7: 4 2 1 is the stairs 3 2 1 with the extra counter on top of pile 1; 3 3 1 has it on pile 2; 3 2 2 on pile 3; 3 2 1 1 has it as a new pile 4 of height 1. The extra counter moves one pile to the right each move, and from the fourth position jumps back to the first. For 8, the long loop: extras on piles (1, 2) in 4 3 1, then (2, 3) in 3 3 2, then (3, 4) in 3 2 2 1, then (4, 1) in 4 2 1 1, then (1, 2) again; the short loop: (1, 3) in 4 2 2, then (2, 4) in 3 3 1 1, then (1, 3) again. Both extra counters step right together, each wrapping from position 4 to position 1. This is the whole structure of the loops, see the mathematics section.

Problem 12. $5\ 3\ 2\ 1 \rightarrow 4\ 4\ 2\ 1 \rightarrow 4\ 3\ 3\ 1 \rightarrow 4\ 3\ 2\ 2 \rightarrow 4\ 3\ 2\ 1\ 1 \rightarrow 5\ 3\ 2\ 1$; and 4 3 2 1 1 is already in this loop. Five arrangements: the extra counter on piles 1, 2, 3, 4, then as a fifth pile, then back.

Problem 13. From 10: 9 1, 8 2, 7 2 1, 6 3 1, 5 3 2, 4 3 2 1, 6 moves. More than 6: for instance 6 2 2 (7), 7 3 (8), 4 3 3 (9), 8 1 1 (9). More than 9: 5 4 1 (10), 4 4 1 1 (10), 6 2 1 1 (11), 5 2 2 1 (11), 5 2 1 1 1 (11). The most possible is 12, from exactly three starts: 3 3 2 1 1, 3 2 2 2 1, 3 2 2 1 1 1. The run from 3 3 2 1 1: 5 2 2 1, 4 4 1 1, 4 3 3, 3 3 2 2, 4 2 2 1 1, 5 3 1 1, 4 4 2, 3 3 3 1, 4 2 2 2, 4 3 1 1 1, 5 3 2, 4 3 2 1. Every start is in the reference table for 10 counters with its number of moves, so you can judge any claim at once.

Problem 14. With 3 counters: 3 takes 1 move, 2 1 takes 0, 1 1 1 takes 2, so the most is 2. The most moves: 2 for 3 counters, 6 for 6, 12 for 10, and 20 for 15. The pattern: $1 \cdot 2, 2 \cdot 3, 3 \cdot 4, 4 \cdot 5$, that is $k(k-1)$ where k is the number of steps of the stairs; the differences 4, 6, 8 also give 20. From 4 4 3 2 1 1 the run is: 6 3 3 2 1, 5 5 2 2 1, 5 4 4 1 1, 5 4 3 3, 4 4 3 2 2, 5 3 3 2 1 1, 6 4 2 2 1, 5 5 3 1 1, 5 4 4 2, 4 4 3 3 1, 5 3 3 2 2, 5 4 2 2 1 1, 6 4 3 1 1, 5 5 3 2, 4 4 4 2 1, 5 3 3 3 1, 5 4 2 2 2, 5 4 3 1 1 1, 6 4 3 2, 5 4 3 2 1: 20 moves, and no start with 15 counters takes more. (That no start ever needs more than $k(k-1)$ moves is a theorem of Igusa and Etienne; see the mathematics section.)

Hints, in the order to give them. (1) For Problem 3: “Fix the biggest pile first.” (2) For Problem 5: “Which column must the arrangement after one move be in?” (one column to the right). (3) For Problem 7(b): “Can an old pile still be as big as the old biggest pile?” (4) For Problem 7(d): “Line up the sizes after the move from biggest to smallest; now line up the sizes before. They are the same list.” (5) For Problem 8(c): “How many counters are in the new pile? How many piles can there be after a move?” (6) For Problem 11: “Color the stairs 3 2 1 first in every picture; then look only at the other color.”

If the table stalls. If Problem 7 is too abstract, do it with 6 counters: build 3 3, 4 1 1, 2 2 2 and ask of each “after the move, which is the biggest pile, and how big was the biggest pile before?”; the stop is then the case where the two answers must agree. If the map is messy, redraw it on a blank sheet with the chain from 2 2 1 1 along the bottom.

If the table finishes early. Problems 12, 13, 14 in that order. After that: “12 counters have two loops of five; find both” (5 4 2 1, 4 4 3 1, 4 3 3 2, 4 3 2 2 1, 5 3 2 1 1 and 5 3 3 1, 4 4 2 2, 4 3 3 1 1, 5 3 2 2, 4 4 2 1 1); “how many arrangements of 7 counters does nothing become? of 8?” (5 of 15, 7 of 22, by the test); and the open-ended “predict the length of the loops for 9, 11, 12 counters before running them” (4, 5, 5: the loop length divides the number of steps of the next stairs).

Closing share (eight minutes)

Each table reports for two minutes, youngest first, with counters on the tray.

- K–1: one child builds 3 2 1 and does the move while the other two chant “one from every pile”; the piles come back as 3 2 1. Then a child shows 5 counters going around (3 2, 2 2 1, 3 1 1). The adult asks the children which numbers made perfect stairs (1, 3, 6, 10).
- Grades 2–3: a pair shows the longest run with 6 counters, 2 2 1 1 to 3 2 1 in six moves, and reports that all eleven ways end at 3 2 1; the other pair reports that none of the seven ways with 5 counters is a stop.
- Grades 4–5: one child shows the map; one explains why nothing becomes 2 2 1 1; one explains why a stop with three piles has to be 3 2 1.
- The organizer closes: “So the numbers that have a stop are the stairs numbers 1, 3, 6, 10, 15, 21, and with those numbers every start ends at the stairs. That every start does so is a theorem, proved in 1982 by Jørgen Brandt, and the way the extra counters walk down the stairs is the whole story of the loops.”

The mathematics behind the session (for the two mathematicians)

Setting. A state is a partition $\lambda = (\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p)$ of n , drawn as a Ferrers diagram with λ_i cells in row i (the children draw the transpose, towers of dots; nothing below depends on which). The move B removes the first column (p cells) and inserts it as a row of length p , then re-sorts. This is Bulgarian solitaire (the name is from Martin Gardner’s Scientific American column of August 1983; the game was circulating among mathematicians a few years earlier). The dynamics is a map on the finite set of $p(n)$ partitions, so every orbit is eventually periodic: a transient and then a cycle; a stop is a cycle of length one.

Fixed points. $B(\lambda) = \lambda$ forces λ to be a staircase $(k, k-1, \dots, 1)$, so $n = T_k = k(k+1)/2$. The grades 4–5 Problem 7 is the proof: the multiset $\{p, \lambda_1 - 1, \dots, \lambda_p - 1\}$ (zeros dropped) must equal $\{\lambda_1, \dots, \lambda_p\}$; the largest element of the first multiset must be λ_1 , and it cannot be $\lambda_1 - 1$, so it is p and $\lambda_1 = p$; matching the sorted lists position by position then gives $\lambda_{i+1} = \lambda_i - 1$, and exactly one zero must be dropped, so $\lambda_p = 1$.

Brandt’s theorem (1982). If $n = T_k$, every partition of n reaches the staircase. For other n no fixed point exists, and the cycles are described below. A proof in two steps, in the style of Akin and Davis (1985):

Step 1, a monovariant. Label the cell in row i , column j with its anti-diagonal number $i + j - 1$ and let $W(\lambda)$ be the sum of the labels of all n cells. Perform the move without sorting: delete the first column and put it on top as a new first row. A cell (i, j) with $j \geq 2$ goes to $(i+1, j-1)$, same label; the cell $(i, 1)$ goes to $(1, i)$, same label. So the unsorted move preserves the number of cells on every anti-diagonal, and in particular preserves W . The result has rows $p, \lambda_1 - 1, \lambda_2 - 1, \dots$, which is already sorted when $p \geq \lambda_1 - 1$. If not, the new row must drop to its place below the rows $\lambda_1 - 1 \geq \dots \geq \lambda_{m-1} - 1 > p$; those $m-1$ rows each move up one row (every cell’s label drops by one, and they hold more than $(m-1)p$ cells) while the p cells of the new row each gain $m-1$. Hence W strictly decreases whenever sorting is needed. W is a positive integer, so only finitely many moves need sorting: from some time on, every move is the pure anti-diagonal rotation.

Step 2, rotations. Once no sorting is needed, the cells on anti-diagonal s (positions $(i, s + 1 - i)$, $i = 1, \dots, s$) are simply rotated one step per move, with period s , and the diagram must be a Ferrers diagram at all times. Let k be the largest index such that anti-diagonals $1, \dots, k$ are all full, so diagonal $s = k + 1$ has a gap. Suppose a cell is present on anti-diagonal $s + 1$. A gap on diagonal s at position i forbids positions i and $i + 1$ on diagonal $s + 1$ (the cell to its right and the cell below it). Since $\gcd(s, s + 1) = 1$, after a suitable number of moves the gap sits at position 1 of diagonal s while the cell sits at position 1 of diagonal $s + 1$: the cell $(1, s + 1)$ is present with $(1, s)$ absent, which is not a Ferrers diagram. So diagonal $s + 1$ is empty, and then so is every later one. In a cycle, therefore, diagonals $1, \dots, k$ are full, diagonal $k + 1$ carries c cells with $0 \leq c \leq k$, and nothing else: $n = T_k + c$. If n is triangular this forces $c = 0$: the state is the staircase and the cycle is a fixed point. That is Brandt's theorem.

The cycles for non-triangular n . Write $T_{k-1} < n < T_k$, $c = n - T_{k-1}$. By Step 2 every cycle state is the staircase of order $k - 1$ with c extra cells on the k -th anti-diagonal ($1 \leq c \leq k - 1$), and the move rotates those c cells cyclically among the k positions of that diagonal. The children's "extra counter walking down the stairs" (K-1 Problems 14–15, grades 4–5 Problem 11) is exactly this rotation. Hence: the cycles for n correspond to the binary necklaces of length k with c beads marked, and the length of a cycle is the period of its necklace, a divisor of k . Examples: $n = 4, 5$ ($k = 3$): one cycle of length 3. $n = 7$ ($k = 4$, $c = 1$): one cycle of length 4. $n = 8$ ($k = 4$, $c = 2$): necklaces 1100 (period 4) and 1010 (period 2), two cycles, lengths 4 and 2. $n = 9$: one cycle of length 4. $n = 11$: one cycle of length 5. $n = 12$ ($c = 2$, $k = 5$): two cycles of length 5. The number of cycles for given n is a necklace count, computable by Burnside's lemma.

How long the transient is. Brandt conjectured, and Igusa (1985) and Etienne (1991) proved, that for $n = T_k$ the staircase is reached within $k^2 - k$ moves. The bound is attained for every k we checked by computer ($k \leq 7$): 1 1 1 for $k = 2$ (2 moves), 2 2 1 1 for $k = 3$ (6 moves), 3 3 2 1 1, 3 2 2 2 1, 3 2 2 1 1 1 for $k = 4$ (12 moves), 4 4 3 2 1 1 among six starts for $k = 5$ (20 moves).

Garden of Eden states. μ with p parts has a preimage under B if and only if some part of μ is at least $p - 1$, that is, $\mu_1 \geq p - 1$: the preimage is obtained by deleting one part $m \geq p - 1$, adding 1 to each remaining part and appending $m - p + 1$ parts equal to 1, and conversely the new pile created by a move has size equal to the previous number of parts, which is at least the new number of parts minus one. The number of preimages of μ is the number of distinct part sizes $m \geq p - 1$ (different m give preimages with different numbers of parts). So 3 2 1 has two preimages (4 2 and itself), 4 3 2 has three, and the unreachable states are those with $\mu_1 \leq p - 2$: 1 of 3 partitions for $n = 3$, 3 of 11 for $n = 6$, 5 of 15 for $n = 7$, 7 of 22 for $n = 8$, 14 of 42 for $n = 10$ (all checked by computer, as are all the tables below).

Where it leads. Partitions and Ferrers diagrams: conjugation (the children's towers are the conjugate of the row picture), $p(n)$ and its generating function, the completeness arguments (grades 2–3 Problems 6–7, grades 4–5 Problem 3) as the first steps of counting partitions by largest part. Finite dynamical systems: functional graphs, transients and cycles, monovariants (the standard tool for termination proofs in olympiad combinatorics and in computer science), Garden of Eden states (the name comes from cellular automata, Moore 1962). Necklaces and Burnside's lemma for counting the cycles. Research directions: Akin and Davis (1985) and Griggs and Ho (1998) study the general "shift" maps on partitions and compositions (Austrian and Carolina solitaire); Hopkins and Jones (2006) the cycle structure for compositions; Popov, and Eriksson and Jonsson, random Bulgarian solitaire and its limit shapes; and the whole subject sits next to chip-firing games and the abelian sandpile model (Björner, Lovász and Shor 1991; Dhar 1990), where "take one from every pile" becomes "every vertex that can fire does." Surveys: Hopkins, "30 years of Bulgarian solitaire," College Mathematics Journal 43 (2012); Drensky, "The Bulgarian solitaire and the mathematics around it" (2015).

Reference tables

Each table lists every arrangement of n counters, the arrangement it becomes after one move, the number of moves until the stop or until the loop is entered, and the arrangements that become it after one move (“—” means none: an unreachable start).

4 counters (loop 3 1, 2 2, 2 1 1)

start	next	moves	comes from
4	3 1	1	1 1 1 1
3 1	2 2	0	4, 2 1 1
2 2	2 1 1	0	3 1
2 1 1	3 1	0	2 2
1 1 1 1	4	2	—

5 counters (loop 3 2, 2 2 1, 3 1 1)

start	next	moves	comes from
5	4 1	2	1 1 1 1 1
4 1	3 2	1	5, 2 1 1 1
3 2	2 2 1	0	4 1, 3 1 1
3 1 1	3 2	0	2 2 1
2 2 1	3 1 1	0	3 2
2 1 1 1	4 1	2	—
1 1 1 1 1	5	3	—

6 counters (stop 3 2 1)

start	next	moves	comes from
6	5 1	3	1 1 1 1 1 1
5 1	4 2	2	6, 2 1 1 1 1
4 2	3 2 1	1	5 1, 3 1 1 1
4 1 1	3 3	5	2 2 1 1
3 3	2 2 2	4	4 1 1
3 2 1	3 2 1	0	4 2, 3 2 1
3 1 1 1	4 2	2	2 2 2
2 2 2	3 1 1 1	3	3 3
2 2 1 1	4 1 1	6	—
2 1 1 1 1	5 1	3	—
1 1 1 1 1 1	6	4	—

7 counters (loop 4 2 1, 3 3 1, 3 2 2, 3 2 1 1)

start	next	moves	comes from
7	6 1	3	1 1 1 1 1 1 1
6 1	5 2	2	7, 2 1 1 1 1 1
5 2	4 2 1	1	6 1, 3 1 1 1 1
5 1 1	4 3	2	2 2 1 1 1
4 3	3 2 2	1	5 1 1, 4 1 1 1
4 2 1	3 3 1	0	5 2, 3 2 1 1
4 1 1 1	4 3	2	2 2 2 1
3 3 1	3 2 2	0	4 2 1
3 2 2	3 2 1 1	0	4 3, 3 3 1
3 2 1 1	4 2 1	0	3 2 2
3 1 1 1 1	5 2	2	—
2 2 2 1	4 1 1 1	3	—
2 2 1 1 1	5 1 1	3	—
2 1 1 1 1 1	6 1	3	—
1 1 1 1 1 1 1	7	4	—

8 counters (loops 4 3 1, 3 3 2, 3 2 2 1, 4 2 1 1 and 4 2 2, 3 3 1 1)

start	next	moves	comes from
8	7 1	4	1 1 1 1 1 1 1 1
7 1	6 2	3	8, 2 1 1 1 1 1 1
6 2	5 2 1	2	7 1, 3 1 1 1 1 1
6 1 1	5 3	2	2 2 1 1 1 1
5 3	4 2 2	1	6 1 1, 4 1 1 1 1
5 2 1	4 3 1	1	6 2, 3 2 1 1 1
5 1 1 1	4 4	2	2 2 2 1 1
4 4	3 3 2	1	5 1 1 1
4 3 1	3 3 2	0	5 2 1, 4 2 1 1
4 2 2	3 3 1 1	0	5 3, 3 3 1 1
4 2 1 1	4 3 1	0	3 2 2 1
4 1 1 1 1	5 3	2	2 2 2 2
3 3 2	3 2 2 1	0	4 4, 4 3 1
3 3 1 1	4 2 2	0	4 2 2
3 2 2 1	4 2 1 1	0	3 3 2
3 2 1 1 1	5 2 1	2	—
3 1 1 1 1 1	6 2	3	—
2 2 2 2	4 1 1 1 1	3	—
2 2 2 1 1	5 1 1 1	3	—
2 2 1 1 1 1	6 1 1	3	—
2 1 1 1 1 1 1	7 1	4	—
1 1 1 1 1 1 1 1	8	5	—

10 counters (stop 4 3 2 1; “moves” is the number of moves to the stop; the unreachable starts are those whose biggest pile is at most the number of piles minus 2)

start	next	moves
10	9 1	6
9 1	8 2	5
8 2	7 2 1	4
8 1 1	7 3	9
7 3	6 2 2	8
7 2 1	6 3 1	3
7 1 1 1	6 4	3
6 4	5 3 2	2
6 3 1	5 3 2	2
6 2 2	5 3 1 1	7
6 2 1 1	5 4 1	11
6 1 1 1 1	5 5	7
5 5	4 4 2	6
5 4 1	4 3 3	10

start	next	moves
5 3 2	4 3 2 1	1
5 3 1 1	4 4 2	6
5 2 2 1	4 4 1 1	11
5 2 1 1 1	5 4 1	11
5 1 1 1 1 1	6 4	3
4 4 2	3 3 3 1	5
4 4 1 1	4 3 3	10
4 3 3	3 3 2 2	9
4 3 2 1	4 3 2 1	0
4 3 1 1 1	5 3 2	2
4 2 2 2	4 3 1 1 1	3
4 2 2 1 1	5 3 1 1	7
4 2 1 1 1 1	6 3 1	3
4 1 1 1 1 1 1	7 3	9

start	next	moves
3 3 3 1	4 2 2 2	4
3 3 2 2	4 2 2 1 1	8
3 3 2 1 1	5 2 2 1	12
3 3 1 1 1 1	6 2 2	8
3 2 2 2 1	5 2 1 1 1	12
3 2 2 1 1 1	6 2 1 1	12
3 2 1 1 1 1 1	7 2 1	4
3 1 1 1 1 1 1 1	8 2	5
2 2 2 2 2	5 1 1 1 1 1	4
2 2 2 2 1 1	6 1 1 1 1	8
2 2 2 1 1 1 1	7 1 1 1	4
2 2 1 1 1 1 1 1	8 1 1	10
2 1 1 1 1 1 1 1 1	9 1	6
1 1 1 1 1 1 1 1 1 1	10	7